From the above equations,

$$W_2 = \frac{15}{16}E^2C \tag{16}$$

5.

1. **Particle speeds on entering the magnetic field**

At $t = 0$ open D_1. Since $u = U_0\sin\left(\frac{2\pi}{T}t\right)$, C_1's voltage is $u = 0$, so all particles pass C_1. After C_1, if they reach C_2 at $t = T/2$ or integer multiples of $T/2$, C_2's voltage is zero, so they pass C_2. To pass C_1 then C_2, their speeds v_n must satisfy

$$\frac{l}{v_n} = n\frac{T}{2} \quad n = 1,2,\cdots \tag{1}$$

For particles passing C_1 at $t = 0$ then C_2, speeds are

$$v_n = \frac{2l}{nT} \quad n = 1,2,\cdots \tag{2}$$

Time from C_1 to C_2 at speed v_n:

$$\Delta t_{n1} = \frac{1}{2}nT \tag{3}$$

Time from C_2 to G:

$$\Delta t_{n2} = \frac{l}{2v_n} \tag{4}$$

Thus arrival time at G is

$$t_n = \Delta t_{n1} + \Delta t_{n2} = \frac{3}{4}nT \tag{5}$$

Since D_2 shuts at $t = 10T/4$, only if

$$t_n \leqslant \frac{10}{4}T \tag{6}$$

they pass C into the field. Using (5)、(6), n is an integer, so the largest n is

$$n_{max} = 3 \tag{7}$$

Their energies on entering the magnetic field are

$$E_n = \frac{1}{2}mv_n^2 + E_0\sin\left[\frac{2\pi}{T}\left(\frac{3n}{4}T\right) + \frac{\pi}{4}\right]$$
$$n = 1,2,3 \tag{8}$$

Their energies are, respectively,